More about Service Levels

Keystone

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More about Service Levels

Quality of Service (QoS) in Keystone

Storage quality of service (QoS) is a critical technology that ensures that applications obtain consistent and predictable performance. Without QoS, certain workloads, such as those for booting of multiple systems, may consume most or all of the resources for a period, and affect other workloads. For information about QoS, see Guarantee throughput with QoS overview.

Adaptive QoS

Adaptive QoS (AQoS) is used by Keystone services to dynamically maintain the IOPS/TiB ratio based on the volume size. For information about AQoS policies, see About adaptive QoS.

Keystone provides you with AQoS polices that you can set up once your cluster is in production. You should ensure that all your volumes are associated with the correct AQoS policies that are already created and available in your system.

An ONTAP volume is non-compliant if it does not have an AQoS policy applied. A volume without a QoS policy is the last on the list of priority for the system to provide any available input-output operations. However, if any input-output operations are available, then the volume could consume all available IOs.

> If you have not applied AQoS policies to your volumes, those volumes will be measured and charged at the highest service level as per your subscription. This may result in unintended burst charges.

Adaptive QoS settings

The Adaptive QoS (AQoS) settings vary with service levels.

Policy name	Extreme	Premium	Performance	Standard	Value
Expected IOPS	6,144	2,048	1,024	256	64
Expected IOPS Allocation	Allocated space				
Peak IOPS	12,288	4,096	2,048	512	128
Peak IOPS Allocation	Used space				
Block Size	32K				

Configuration of adaptive QoS policy group

You can configure adaptive QoS (AQoS) policies to automatically scale a throughput ceiling or floor to volume size. Not all Keystone service levels are aligned with the default ONTAP QoS policies. You can create custom QoS policies for them. For configuring a policy, you should be aware of the following:

- **Policy group name**: The name of the AQoS policy group. For example, `Keystone_extreme`.

- **VServer**: The name of the VServer or storage VM (storage virtual machine).
- **Expected IOPS**: The minimum number of IOPS, per allocated TiB per volume, that the system attempts to provide when enough system IOPS are available.
- **Peak IOPS**: The maximum number of IOPS, per used TiB per volume, that the system allows the volume to reach before it throttles the IOPS through injection of latency.
- **Expected IOPS allocation**: This parameter controls whether the expected IOPS available to the volume is based on the allocated or used size of the volume. In Keystone, this is based on the allocated space.
- **Peak IOPS allocation**: This parameter controls whether the peak IOPS available to the volume is based on the allocated or used size of the volume. In Keystone, this is based on the used space.
- **Absolute minimum IOPS**: The lowest number of expected IOPS that will be applied to a volume if the volume size is very small and would otherwise result in an unacceptable number of IOPS. This value defaults to 1,000 for `Extreme`, 500 for `Premium`, and 250 for `Performance`, and 75 for `Standard` and `Value` service levels.

> This is not IOPS density (for example, 75 IOPS/TiB), but an absolute minimum number of IOPS.

For information about IO density, see Metrics and definitions used in Keystone Services. For more information about AQoS policy groups, see Use adaptive QoS policy groups.

Settings of adaptive QoS policies

The settings for adaptive QoS (AQoS) policies for each service level are described in the following sections. The minimum and maximum volume sizes for each service level provided here allow for optimal IOPs and latency values for a volume. Creating too many volumes outside of these guidelines may negatively impact performance in those volumes.

Settings for Extreme service level

Settings and commands for the Extreme service level:

- Sample command:

```
qos adaptive-policy-group create -policy-group <Keystone_extreme> -vserver
<SVM_name> -expected-iops 6144 -peak-iops 12288 -expected-iops-allocation
allocated-space -peak-iops-allocation used-space -block-size 32K -absolute
-min-iops 1000
```

- Minimum volume size: 100GiB, 0.1TiB
- Maximum volume size: 10TiB

Settings for Premium service level

Settings and commands for the Premium service level:

- Sample command:

```
qos adaptive-policy-group create -policy-group <Keystone_premium> -vserver
<SVM_name> -expected-iops 2048 -peak-iops 4096 -expected-iops-allocation
allocated-space -peak-iops-allocation used-space -block-size 32K -absolute
-min-iops 500
```

- Minimum volume size: 500GiB, 0.5TiB
- Maximum volume size: 50TiB

Settings for Performance service level

Settings and commands for the Performance service level:

- Sample command:

```
qos adaptive-policy-group create -policy-group <Keystone_performance>
-vserver <SVM_name> -expected-iops 1024 -peak-iops 2048 -expected-iops
-allocation allocated-space -peak-iops-allocation used-space -block-size
32K -absolute-min-iops 250
```

- Minimum volume size: 500GiB, 0.5TiB
- Maximum volume size: 80TiB

Settings for Standard service level

Settings and commands for the Standard service level:

- Sample command:

```
qos adaptive-policy-group create -policy-group <Keystone_standard>
-vserver <SVM_name> -expected-iops 256 -peak-iops 512 -expected-iops
-allocation allocated-space -peak-iops-allocation used-space -block-size
32K -absolute-min-iops 75
```

- Minimum volume size: 1TiB
- Maximum volume size: 100TiB

Settings for Value service level

Settings and commands for the Value service level:

- Sample command:

```
qos adaptive-policy-group create -policy-group <Keystone_value> -vserver
<SVM_name> -expected-iops 64 -peak-iops 128 -expected-iops-allocation
allocated-space -peak-iops-allocation used-space -block-size 32K -absolute
-min-iops 75
```

- Minimum volume size: 1TiB
- Maximum volume size: 100TiB

Block size calculation

Note these points before you calculate the block size by using these settings:

- IOPS/TiB = MBps/TiB divided by (block size * 1024)
- Block size is in KB/IO
- TiB = 1024GiB; GiB = 1024MiB; MiB = 1024KiB; KiB = 1024Bytes; as per base 2
- TB = 1000GB; GB = 1000MB; MB = 1000KB; KB = 1000Bytes; as per base 10

Sample block size calculation

To calculate the throughput for a service level, for example `Extreme` service level:

- Maximum IOPS: 12,288
- Block size per I/O: 32KB
- Maximum throughput = (12288 * 32 * 1024) / (1024*1024) = 384MBps/TiB

If a volume has 700GiB of logical used data, the available throughput will be:

```
Maximum throughput = 384 * 0.7 = 268.8MBps
```

Capacity requirements for service levels

The capacity requirements for Keystone STaaS service levels differ with the file, block, object, or cloud storage supported by the Keystone STaaS subscription.

Minimum capacity requirements for file and block services

The minimum capacity and incremental capacity allowed per subscription is described in the following table. The minimum capacity per service level is defined to be the same across Keystone sales motions. The capacity above the minimum capacity either at the beginning of the subscription, or as an add-on service to the subscription, or after reallocation during the subscription is also structured in the table.

Capacity	Extreme	Premium	Performance	Standard	Value
Minimum capacity [in TiB]	25			100	

Incremental capacity (and in multiples) allowed at start of subscription [in TiB]	25	25
Incremental capacity (and in multiples) allowed as add-on during subscription [in TiB]	25	25

Minimum capacity requirements for object storage

You can see the minimum capacity requirements for object storage in the following table:

Capacity	Data tiering	Object	Cloud Volumes ONTAP	Cloud Backup service
Minimum capacity [in TiB]	Not applicable	500	4	4
Incremental capacity (and in multiples) allowed at start of subscription [in TiB]	Not applicable	100	1	1
Incremental capacity (and in multiples) allowed as add-on during subscription [in TiB]	Not applicable	100	1	1

Capacity adjustments

Note the following clauses for capacity adjustments:

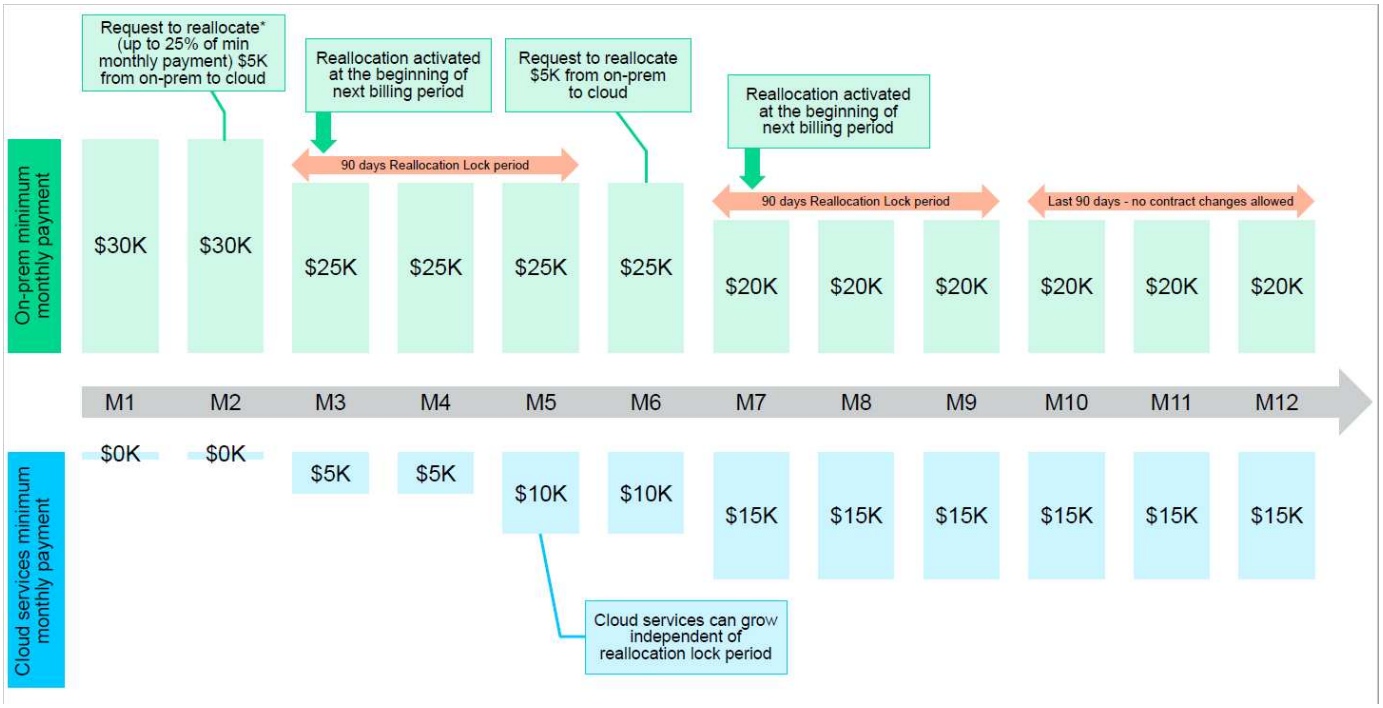
- Capacity can be added anytime during the term, except for the last 90 days of the contract term, in the increments per service level as described in the tables in the previous section. Addition of capacity and/or services is allowed within the last 90 days of the contract term as long as there is a consent of service renewal. Any addition in capacity, new service on-prem or cloud can co-term with the existing term. The invoice sent to you following the activation of the new services reflects the revised billing. Committed capacity of cloud services cannot be reduced at any point during the subscription term. Meanwhile, committed capacity and committed spend on the on-premises services during the term of the contract can be reduced based on certain criteria as defined in the following section *Capacity reduction*.
- A burst capacity is available at each site, based on the Keystone agreement. Usually, it is 20% above the committed capacity for a service level. Any burst usage is billed only for that billing period. If you have

additional burst requirement greater than the capacity you agreed upon, contact support.

- Committed capacity can be altered during a contract term, only under certain conditions, as described in the following section *Capacity reduction*.
- Increasing capacity or changing to higher service level during a subscription term is allowed. However, moving from a higher service level to a lower service level is not permitted.
- Any change request in the last 90 days of the service term requires a renewal of the service for a minimum of one year.

Capacity reduction

Capacity reduction (annual) is applicable to the *Annual in Advance* payment model and on-premises only deployments. It is not available for cloud services or hybrid cloud services. It provides provision for on-premises capacity, which can be reduced by up to 25% per service level per subscription. This reduction is allowed once every year to be made effective at the beginning of the next annual billing period. On-premises service-based annual payments should be >=$200K anytime during the term in order to take advantage of capacity reduction. Because it is supported only for on-premises deployments, this billing model does not provide reallocation in spending from on-premises to cloud services. An example of annual capacity reduction is illustrated in the following image.



Quarterly spend reallocation

Keystone STaaS offers you the option to reallocate on-premises service spend to Cloud Volumes ONTAP spend.

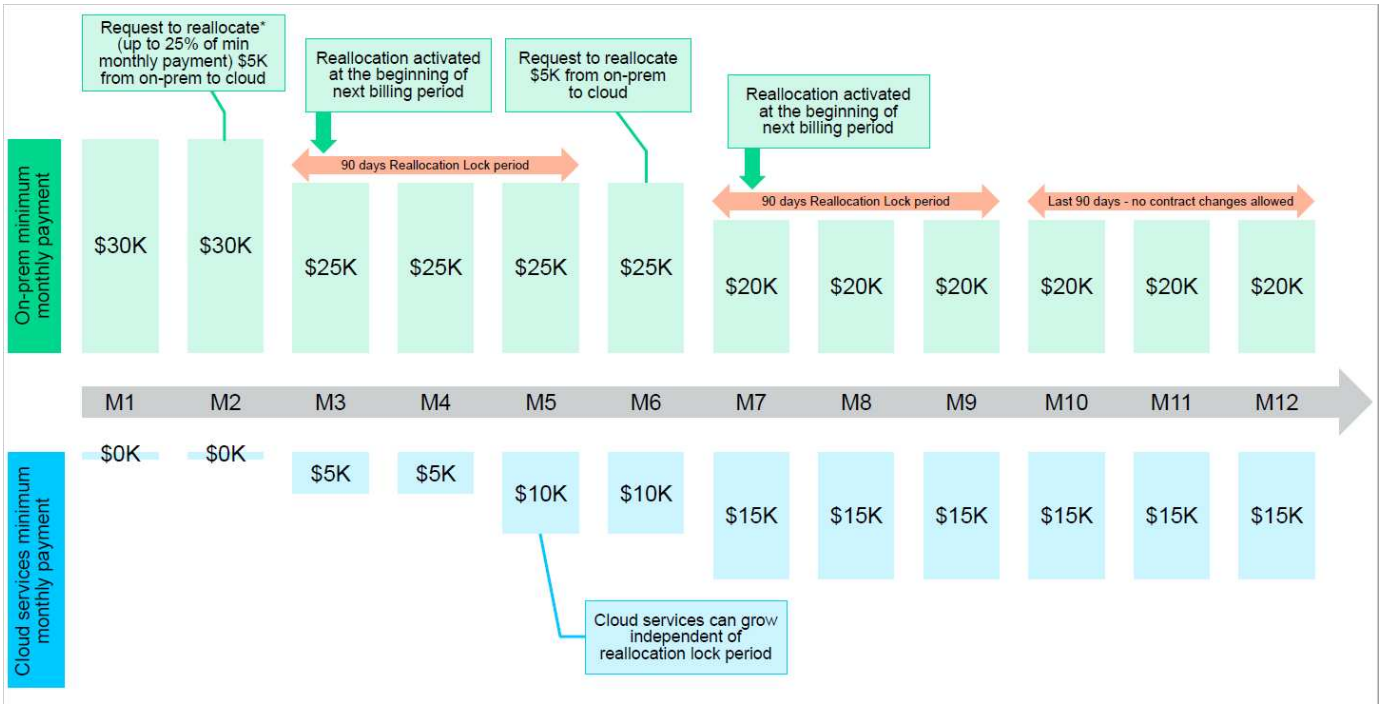
Requirements and conditions at a subscription level:

- Applies only to monthly billing in arrear model.
- Applies only to subscriptions with 1, 2, or 3-year term commitments.
- Capacity for Cloud Volumes ONTAP and Cloud Backup service should be purchased through Keystone.
- Up to 25% of the existing on-premises, service-based monthly payments can be used for reallocation to

cloud services.

- Reallocation requests are made effective only after 90 days from the previous activation date of the reallocation.
- Reallocation cannot be done from cloud services back to on-premises services.
- A request to reallocate should be formally submitted by the customer or partner to Keystone Success Manager (KSM) at least one week before the next billing cycle.
- New requests go into effect only from the consecutive billing cycle.

You can allocate a portion of your expenses towards your subscribed file, block, or object storage service levels to hybrid cloud storage services. Up to 25% of the Annual Contract Value (ACV) can be reallocated on a quarterly basis to Cloud Volumes ONTAP Primary and Cloud Volumes ONTAP Secondary services:



This table provides a set of sample values to demonstrate how the reallocation of expenses works. In this example, `$5000` from the monthly spend is reallocated to hybrid cloud storage service.

Before allocation	**Capacity (TiB)**	**Monthly designated expense**
Extreme	125	37,376
After reallocation	**Capacity (TiB)**	**Monthly designated expense**
Extreme	108	37,376
Cloud Volumes ONTAP	47	5,000
		37,376

The reduction is of (125-108) = 17 TiB of the capacity allocated for the Extreme service level. On spend reallocation, the allotted hybrid cloud storage is not of 17 TiB but an equivalent capacity that $5000 can purchase. In this example, for $5000, you can get 17 TiB on-prem storage capacity for the Extreme service level and 47 TiB hybrid cloud capacity of Cloud Volumes ONTAP service level. Therefore, the reallocation is with respect to the spend, not capacity.

Contact your Keystone Success Manager (KSM) if you want to reallocate expenses from your on-premises services to cloud services.

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